

DATA SCIENCE CAPSTONE · LT #1

Lightning Talk #1: Pitch + Problem Statement

Four topics — problem statement, goals, approach, data, and EDA findings for each.



TOPIC 1

Turbofan Engine Pressure Prediction

REGRESSION

Physical sensors on jet engines can degrade, fail, or give noisy readings under extreme conditions. If the HPC outlet static pressure sensor fails mid-flight, can we estimate it from the engine's other readings to keep the aircraft flying safely?

GOAL

- Predict HPC Outlet Static Pressure using other sensors and operating settings
- Build a “virtual sensor” regression model to act as a fail-safe backup
- Identify which engine metrics (temps, speeds) matter most

APPROACH

- Engineer features: rolling 5-cycle sensor averages, engine wear-stage bins, compressor thermal ratio
- Standardize sensor units ($^{\circ}\text{R} \rightarrow ^{\circ}\text{C}$, psia $\rightarrow$ bar) before modeling
- Baseline Linear Regression, then Random Forest

EDA

DATA SOURCE

NASA C-MAPSS Turbofan Degradation Simulation (FD001, Kaggle)

SIZE

20,631 rows $\times$ 26 columns (raw)

MISSING VALUES

None found (0 nulls across all columns)

FEATURE SPLIT

31 numerical, 1 categorical (after feature engineering)

OUTLIERS

Extreme values likely reflect valid operating states — plan to cap via a robust scaler rather than remove



TOPIC 2

Steam Game Popularity Prediction

REGRESSION

Tens of thousands of games release on Steam each year — most never gain traction, while a few capture massive, highly engaged audiences. Can peak concurrent players be predicted from a game's release metadata?

GOAL

- Predict a game's Peak CCU from pricing, platform, and genre metadata
- Identify which structured factors most strongly correlate with high engagement

APPROACH

- Log-transform Peak CCU ($\log_{10}$) to handle its extreme right skew
- Drop near-empty / unstructured text columns; language, category & tag counts
- Baseline Linear Regression, then Random Forest / XGBoost

EDA

DATA SOURCE

Steam Games Dataset (Kaggle)

SIZE

125,855 rows $\times$ 39 cols $\rightarrow$ cleaned to 125,854 $\times$ 29

MISSING VALUES

Heavy in text/URL fields (e.g. Reviews 90%, Website 60%) — dropped or imputed; 0 remain after cleaning

FEATURE SPLIT

18 numerical, 3 boolean platform flags

OUTLIERS

Price capped at the 99th percentile (collector's editions / bundles)



TOPIC 3

Aviation Accident Severity Prediction

CLASSIFICATION

Given the circumstances of a flight — aircraft type, engine type, weather, phase of flight — can we predict accident injury severity to help identify higher-risk conditions?

GOAL

- Classify accident injury severity from structured incident data
- Surface which operational/mechanical factors most strongly predict severe outcomes

APPROACH

- Use only structured fields; exclude free-text narrative/remarks
- Clean messy target labels (e.g. “Fatal(1)”, “Fatal(10)”) into Fatal / Serious / Minor / Incident
- Random Forest classifier with class-weighting

EDA

DATA SOURCE

NTSB Aviation Accident Database, 1962–2023 (Kaggle)

SIZE

88,889 rows × 31 columns

MISSING VALUES

Substantial in several fields (Aircraft.Category ~64%, Schedule ~86%) — impute “Unknown” or drop sparse ID columns

FEATURE SPLIT

5 numerical, 26 categorical

OUTLIERS

Number of Engines has implausible values (0 or very high)



TOPIC 4

Weather Temperature Prediction

REGRESSION

Weather conditions affect everything from outdoor equipment performance to flight safety. Can air temperature be predicted from other atmospheric measurements?

GOAL

- Predict temperature (°C) from humidity, wind, pressure & visibility
- Quantify which atmospheric variables most strongly drive temperature

APPROACH

- Drop constant “Loud Cover” column and duplicate rows
- Exclude Apparent Temperature as a feature — it correlates 0.99 with the target and would leak the answer
- Baseline Linear Regression, then Random Forest

EDA

DATA SOURCE

Weather in Szeged, Hungary, 2006–2016 (Kaggle)

SIZE

96.5k rows × 12 cols → cleaned to 96,429 × 11

MISSING VALUES

517 nulls in Precip Type; all other columns complete

FEATURE SPLIT

7 numerical, 4 categorical

OUTLIERS

Humidity/Pressure = 0 treated as sensor error; high wind speeds capped, not dropped

RECAP

Four Topics Summary

Topic	Type	Rows	Target
Turbofan Engine Pressure	Regression	20,631	HPC Outlet Static Pressure
Steam Game Popularity	Regression	125,855	Peak CCU (log-transformed)
Aviation Accident Severity	Classification	88,889	Injury Severity
Weather Temperature	Regression	96,429	Temperature (°C)

REMINDERS

- Each dataset needs 15k+ rows and 10+ columns — all four topics
- No text-based datasets — all four use structured/tabular data